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Mesh4D: 4D Mesh Reconstruction and Tracking from Monocular Video

Zeren Jiang[1] Chuanxia Zheng[1,2] Iro Laina[1] Diane Larlus[3] Andrea Vedaldi[1]
[1]VGG, University of Oxford [2]Nanyang Technological University [3]Naver Labs Europe
{zeren, cxzheng, iro, vedaldi}@robots.ox.ac.uk diane.larlus@naverlabs.com

mesh-4d.github.io

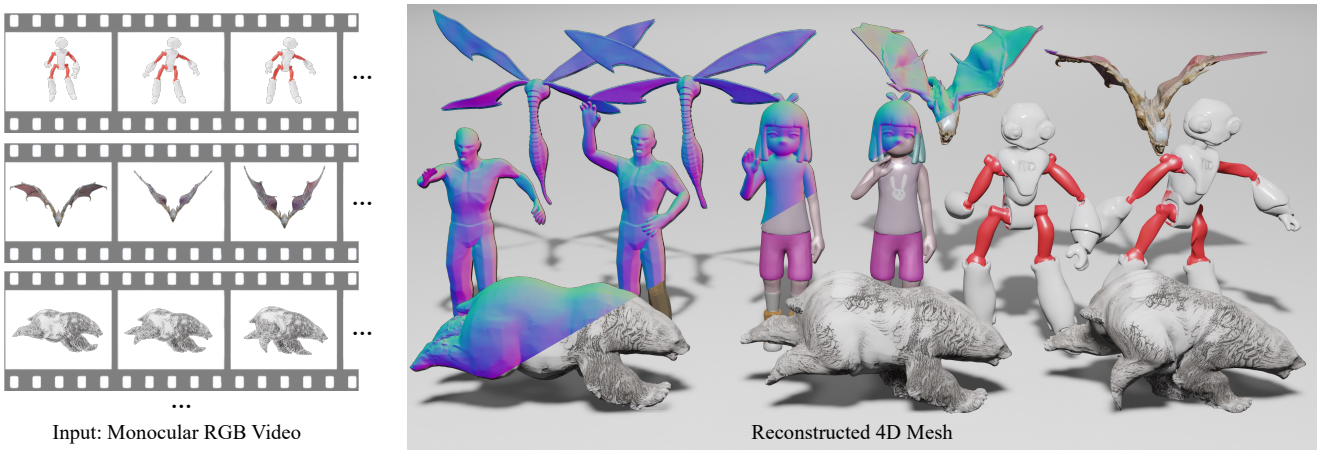

Figure 1. **Illustration of Mesh4D**. Given a monocular RGB video as input, Mesh4D generates a complete animated 3D mesh and its deformation. Each 4D reconstruction is shown at several time steps, the top layer displaying normals and the bottom one textured meshes.

Abstract

We propose Mesh4D, a feed-forward model for monocular 4D mesh reconstruction. Given a monocular video of a dynamic object, our model reconstructs the object's complete 3D shape and motion, represented as a deformation field. Our key contribution is a compact latent space that encodes the entire animation sequence. This latent space is learned by an autoencoder that, during training, is guided by the skeletal structure of the training objects, providing strong priors on plausible deformations. Crucially, skeletal information is not required at inference time. The encoder employs spatio-temporal attention, yielding a more stable representation of the object's overall deformation. Building on this representation, we train a latent diffusion model that, conditioned on the input video and the mesh reconstructed from the first frame, predicts the full animation in one shot. We evaluate Mesh4D on reconstruction and novel view synthesis benchmarks, outperforming prior methods in recovering accurate 3D shape and deformation.

1. Introduction

In *monocular 4D shape reconstruction*, we are interested in reconstructing the 3D shape and motion of a dynamic object captured in a monocular RGB video. 4D reconstruction has many applications in vision, graphics, and robotics. For example, it can be used to automate the creation of 3D animations, which is costly and time-consuming to do by hand, and requires domain expertise. However, 4D reconstruction is very challenging as the monocular video contains limited information about the 3D shape and motion of the object, which instead must be recovered in full, beyond what is visible in the video.

Often [16, 35, 77] this problem has been tackled using analysis-by-synthesis, where an animated 3D shape is optimized iteratively to fit the input RGB(D) frames. However, these methods can only reconstruct the *visible portion* of the object and often produce noisy results due to occlusions and ambiguity. Recent feed-forward methods [11, 19, 44] have emerged as promising alternatives. These networks learn to reconstruct in 3D a frame of a monocular video, as

well as the scene motion between that frame and another. However, each application of the network only recovers the 3D points that are visible, and their motion only between the two frames rather than across the whole sequence. As a result, they are insufficient to recover the complete 4D structure of an animated object.

Reconstructing and tracking the parts of the geometry that are not visible in the video requires strong 3D and physical priors that can only be learned from data. This calls for latent 3D generative models, which have been shown to capture rich priors on 3D shapes [20, 46, 62], at least in the static case. In this paper, we thus ask whether *such generative models can be extended to solve 4D reconstruction*, implicitly capturing cues such as symmetry and smoothness that can help to recover the 3D shape and its motion beyond what is directly visible in the video. Recent contributions such as GVFD [73] have already explored this idea and introduced versions of 3D generators [46, 62] that can handle dynamics. However, their ultimate aim is to generate good-looking images of the object and they are less concerned with accurately recovering the 3D shape and motion.

In this work, we introduce Mesh4D, a feed-forward model that learns to accurately reconstruct *3D objects and their motion* from a monocular RGB video (see illustration Fig. 1). We represent the object's shape as a 3D mesh extracted from the first frame of the input video, and then generate a *deformation field* that displaces the surface of the mesh to represent the deformation of the object over time. This representation is intuitive as it factorizes the 3D shape and motion. By comparison, others [18, 41, 54, 69] have approached 4D reconstruction by outputting *independent* 3D reconstructions of each frame and thus do not explicitly model the object's motion. Explicitly modeling motion is often crucial, for example, to texture the 3D object consistently over time.

We argue that, in order to accurately recover the motion of objects, animations should be modeled holistically from the beginning to the end of the video. To make this feasible, we introduce a new Variational Auto-Encoder (VAE) to encode the shape deformation in a compact latent space. Prior work like Motion2VecSet [5] has proposed analogous encodings, but only for two frames at once. In contrast, we show that a latent space representing the motion of the object throughout the whole sequence performs better. To this end, we build a transformer [50] encoder that uses spatial, temporal, and global attention modules [14, 51] in each block, capturing long-term spation-temporal correlations among points.

We also argue for using motion-related information to guide VAE training. In particular, we propose using the object's *skeleton*, as a cue on the space of possible deformations of the object. As shown in Fig. 2, we leverage both skinning weights and bone information as additional inputs to the encoder, which significantly improves the quality of the learned latent space. Importantly, the skeleton is privileged information which is not needed at inference time.

Another challenge in monocular 4D mesh reconstruction is the lack of suitable 4D datasets for training and benchmarking. Following recent works [18, 20, 32, 65], we do not train our deformation model from scratch, but start from a *static* 3D generator [46]. Because the latter is pretrained on large datasets of 3D objects, it helps to achieve robustness and generalization to a wide variety of object types. We then use a dataset of animated 3D objects (with annotated 3D tracks) for training and evaluation. As there are few such datasets, we build one by filtering 3D animated assets from Objaverse [9], selecting high-quality dynamic 3D meshes with skeletons. For evaluation, we propose a new benchmark focusing on the quality of 3D shape and motion reconstruction rather than just the visual quality of the rendered video, which has been the focus of some prior work [5]. To do so, we collect 50 animated 3D assets with significant object motion, high-quality textures, and no overlap with our training data.

To summarize, our contributions are as follows. First, we propose a new model, Mesh4D, for monocular 4D mesh reconstruction. This model reconstructs the 3D shape and deformation of an object from a monocular video. Second, we build our model on top of a new VAE that encodes the object's deformation from the beginning to the end of the sequence in a compact latent space, suitable for latent diffusion. We then train a diffusion model to generate the motion of the object observer in the input video. Third, we propose a benchmark to assess monocular 4D shape reconstruction models with a focus on 3D reconstruction quality instead of 2D rendering quality.

2. Related work

2.1. Optimization-based 4D reconstruction

Iterative or optimization-based methods reconstruct from monocular or multi-view videos by iteratively fitting a 4D representation to them [4, 10, 12, 25, 26, 35, 37, 39, 48, 53, 56, 58, 67, 68]. With the advent of neural radiance fields (NeRFs) [33] and 3D Gaussian Splatting (3D-GS) [21], many time-dependent NeRFs [4, 10, 12, 25, 26, 37, 39] and dynamic 3D Gaussian Splatting methods [53, 56, 58, 67, 68] have been proposed for 4D reconstruction. Although significant progress has been made, most approaches are evaluated on simple scenarios with quasi-static scenes [37, 58] and relatively small datasets [13, 22, 70].

To demonstrate effectiveness on diverse scenarios, recent works explore priors from large-scale pretrained models. One popular direction is to generate multi-view videos with pretrained video diffusion models [1–3, 15] and then perform 4D reconstruction via per-scene optimization [59, 63,

71]. Another line of work [8, 17, 27, 40, 74] directly distills 4D knowledge from large-scale diffusion models using slow score-distillation sampling iterations [38]. However, these methods focus on "plausible" novel view synthesis (NVS), rather than accurate geometry and tracking. More recently, feed-forward 3D generators, such as Hunyuan3D [45–47], TripoSG [24], Trellis [62], and Step1X-3D [23], have shown impressive results on diverse and complex scenarios, but they focus only on static 3D reconstruction. Building upon these feed-forward 3D generators, V2M4 [6] reconstructs 4D meshes from monocular videos by generating per-frame 3D meshes independently, followed by an optimization step to ensure temporal coherence. However, they still require per-scene optimization to achieve consistent 4D reconstruction, which is time-consuming and less flexible. In contrast, our approach directly reconstructs temporally coherent 4D meshes along with tracking from monocular videos in a feed-forward manner. It generalizes to diverse scenarios without per-scene optimization and is therefore much more efficient.

2.2. Feed-forward 4D reconstruction

Like our method, feed-forward 4D reconstruction directly infers a 4D representation from monocular videos in a single pass. MonST3R [75] effectively reconstructs dynamic scenes by adapting the *static* 3D reconstruction produced by DUSt3R [55] with dynamic scene supervision. Numerous follow-ups take a similar path [11, 19, 44], adding functionalities like tracking. However, they mainly work on pairwise frames or stereo videos. To handle long monocular videos, Cut3R [54] introduces a memory bank to store spatial and temporal information, enabling continuous static and dynamic scene reconstruction from a stream of monocular images. π^3 [57] further improves scalability by building a permutation-equivariant architecture on top of VGGT [51]. Beyond DUSt3R-based methods, Geo4D [18] leverages video generators [64] to infer 4D point maps from monocular videos. 4DGT [66] proposes a transformer model to predict dynamic 3D Gaussian representations [21] from monocular videos. However, these approaches mainly focus on visible geometry reconstruction, whereas we address the more challenging task of complete 4D mesh reconstruction with tracking. This is harder due to the need to infer the *invisible* parts of surfaces and establish *dense* correspondences over time.

Some recent works also explore complete 4D reconstruction from monocular videos in a feed-forward manner. L4GM [41] leverages a pretrained ImageDream [52] to generate missing-view images, which are then used to train a feed-forward 4D Gaussian reconstructor. GVFD [73] embeds the 4D mesh data into a Gaussian Variation Field using a compact latent space, and then trains a video-to-4D Gaussian model by inferring latent codes from monocular videos. However, they still focus on producing plausible novel view synthesis, rather than aiming for accurate geometry. More closely related to our task, concurrent work ShapeGen4D [69] encodes a 4D mesh sequence into a sequence of latents and then learns a diffusion model to predict the latents directly, conditioned on monocular videos. However, they align the latents only with a shared set of query points from the first frame and do not directly predict dense correspondences. Instead, our Mesh4D is based on a novel spatio-temporal Transformer architecture operating directly on 4D meshes, which can effectively capture temporal coherence and establish dense correspondences over time. Moreover, compared to Motion2VecSets [5] which processes two meshes at a time, Mesh4D encodes sequences of 4D meshes jointly, and directly predicts complete 4D meshes from monocular videos.

3. Method

Given a monocular video of a moving object, our goal is to reconstruct its 3D shape and motion. Formally, the video $\mathcal{I} = \{\boldsymbol{I}_t\}_{t=1}^{T}$ is a sequence of T RGB images $\boldsymbol{I}_t \in \mathbb{R}^{H\times W\times 3}$. The 3D shape of the object in the first frame of the video is captured by the *3D mesh* $\mathcal{M}_1 = \langle \mathcal{V}_1, \mathcal{F}_1 \rangle$, consisting of an array of 3D vertices $\mathcal{V}_1 \in \mathbb{R}^{N_v\times 3}$ and triangular faces $\mathcal{F}_1 \in \mathbb{N}^{N_f\times 3}$ indexing the vertices. The motion of the object is captured by a dense *deformation field* $\mathcal{T}_{1\to t}$ that specifies the displacement of all points on the mesh (including both vertices and faces) from time 1 to time t. In particular, we can use this deformation field to write the deformed mesh at time t as $\mathcal{M}_t = \langle \mathcal{V}_1 + \mathcal{T}_{1\to t}(\mathcal{V}_1), \mathcal{F}_1 \rangle$.

We cast our problem as learning a neural network Φ that, given the input video, outputs both the *3D shape* and *motion* of the object:

$$\Phi : \mathcal{I} \mapsto \mathcal{M}_1, \{\mathcal{T}_{1\to t}\}_{t=1}^{T}. \tag{1}$$

To ensure that the model generalizes well, especially given the limited availability of 4D training data, we build it on top of a pre-trained image-to-3D generator. We opt for a 3D latent space model due to its robustness and ability to hallucinate complete objects even though only one aspect of them is visible in the image.

Our method has thus three main components. First, it uses an off-the-shelf 'backbone' network for image-to-3D reconstruction, which we introduce in Sec. 3.1. Its purpose is to reconstruct the mesh $\mathcal{M}_1$ from the first frame $\boldsymbol{I}_1$ of the video. Second, it uses a new 4D Variational Auto-Encoder (VAE) to encode the deformation field $\mathcal{T}$ in a compact latent space (Sec. 3.2). Third, it uses a generator that outputs the deformation latent code conditioned on the input video $\mathcal{I}$ and the reference mesh $\mathcal{M}_1$ (Sec. 3.3).

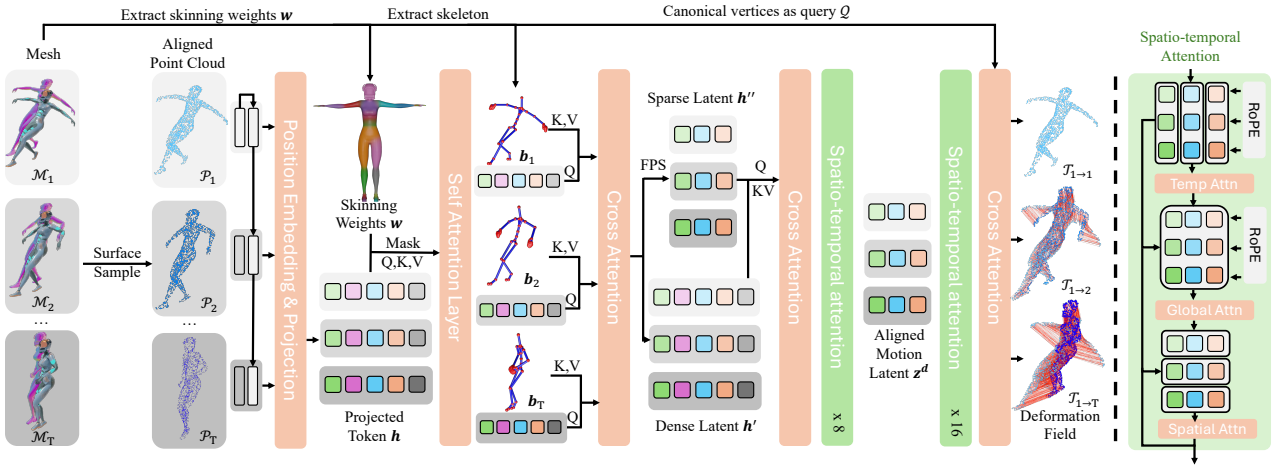

Figure 2. **Overall Deformation VAE pipeline.** (Left) Given a sequence of 3D meshes as input, we uniformly sample a sequence of corresponding points. We inject skeleton information with masked self- and cross-attention. We then apply Farthest Point Sampling (FPS) along the spatial dimension to compress the latent features, followed by eight layers of spatio-temporal attention. The deformation field is decoded by additional spatio-temporal attention layers, followed by cross-attention where canonical vertices serve as query points. (Right) Each spatio-temporal attention layer sequentially performs temporal, global, and spatial attention. For temporal and global attention, we additionally apply 1D RoPE [43] embeddings along the temporal dimension.

3.1. Preliminaries: Latent 3D reconstruction model

Following recent works on 3D diffusion [18, 20, 30, 32], we build Mesh4D on top of a pre-trained latent 3D reconstruction model for *static* objects. We use Hunyuan3D 2.1 [46], an excellent model learned from millions of 3D data samples, but any other similar model could be used instead.

The model uses a VAE to map the 3D mesh $\mathcal{M}$ to a compact latent code $\boldsymbol{z}^s$ based on the VecSet representation [72]. The *encoder* $\boldsymbol{z}^s = \mathcal{E}^s(\mathcal{P}; \boldsymbol{n})$ computes the latents by first uniformly sampling a point cloud $\mathcal{P} \in \mathbb{R}^{M\times 3}$ from the mesh $\mathcal{M}$, augmented with their normals $\boldsymbol{n} \in \mathbb{R}^{M\times 3}$, and then applying a neural network to it. Given the code $\boldsymbol{z}^s$, a *decoder* $\mathcal{D}^s(\mathcal{Q}; \boldsymbol{z}^s)$ recovers the 3D shape of the object by computing its signed distance function (SDF) in correspondence of a set of 3D grid query points $\mathcal{Q} \in \mathbb{R}^{(H\times W\times D)\times 3}$. These are finally converted to a triangle mesh via the marching cubes algorithm [31].

Given the VAE, 3D reconstruction amounts to learning a conditional denoising diffusion generator that can sample the latent code $\boldsymbol{z}^s \sim p(\boldsymbol{z}^s|\boldsymbol{I})$ given the image $\boldsymbol{I}$. This model is trained by minimizing the flow matching objective [29]:

$$\min_{\boldsymbol{\theta}} \mathbb{E}_{(\boldsymbol{z}^s,\boldsymbol{I}),t,\boldsymbol{\epsilon}^s\sim\mathcal{N}(\mathbf{0},\mathbf{1})} \left\|\boldsymbol{v}^s - \boldsymbol{v}^s_{\boldsymbol{\theta}}\left(\boldsymbol{z}^s_t, t, \boldsymbol{I}\right)\right\|_2^2, \quad (2)$$

where $\boldsymbol{z}^s_t = t\boldsymbol{z}^s + (1-t)\boldsymbol{\epsilon}^s$ is the noisy sample at timestep $t \sim \mathcal{U}(0,1)$, and $\boldsymbol{v}^s = \boldsymbol{z}^s - \boldsymbol{\epsilon}$ is the velocity field that moves the noisy sample $\boldsymbol{z}^s_t$ towards the data $\boldsymbol{z}^s$, and $\boldsymbol{v}^s_{\boldsymbol{\theta}}$ denotes the velocity prediction model where $\boldsymbol{\theta}$ are the model parameters. Once trained, one randomly samples the noise from a Gaussian distribution and uses a first-order Euler ordinary differential equation to iteratively transfer the noise $\boldsymbol{\epsilon}^s$ to the data $\hat{\boldsymbol{z}}^s$. Finally, the denoised shape latent $\hat{\boldsymbol{z}}^s$ together with the 3D grid query points $\mathcal{Q}$ are fed into the decoder to output the final mesh.

To obtain a textured mesh, a separate material generation model is used. For this, the input image, rendered multi-view normal maps as well as canonical coordinate maps are taken as condition to generate a PBR texture. Since texturing is orthogonal to our main contribution, we refer the reader to Hunyuan3D 2.1 [46] for more details. Note that this latent 3D reconstruction model is only used to reconstruct the *static* mesh $\mathcal{M}_1$ from the first frame $\boldsymbol{I}_1$, which is used as the reference view for both geometry and texture.

3.2. Deformation variational auto-encoder

In Sec. 3.1, we have obtained the canonical mesh $\mathcal{M}_1 = \langle \mathcal{V}_1, \mathcal{F}_1 \rangle$ from the first frame $\boldsymbol{I}_1$ of the video using an *off-the-shelf* model. Our contribution is to reconstruct the deformation field $\mathcal{T}_{1\to t}$ from $\mathcal{M}_1$ and the video $\mathcal{I}$ as a whole.

Just like the SDF defining the 3D shape of the object, the deformation field $\mathcal{T}_{1\to t}$ is an infinite-dimensional object. The key to our method is to learn a *deformation VAE* that encodes the deformation field into a compact latent space. The encoder $\boldsymbol{z}^d \sim \mathcal{E}^d(\{\mathcal{M}_t\}_{t=1}^T; \boldsymbol{n}, \boldsymbol{w}, \boldsymbol{b})$ maps a sequence of meshes $\{\mathcal{M}_t\}_{t=1}^T$ to a latent code $\boldsymbol{z}^d$, while the decoder takes this latent code and reconstructs the deformation field $\mathcal{T}_{1\to t}(\mathcal{V}_1) = \mathcal{D}^d_t(\mathcal{V}_1; \boldsymbol{z}^d)$ in correspondence of the vertices $\mathcal{V}_1$. The encoder also uses the normal vectors $\boldsymbol{n}$, skinning weights $\boldsymbol{w}$, and bones $\boldsymbol{b}$ to improve the encoding, as we show in Fig. 2 and explain below.

Encoding points in time. Inspired by the design of the VAE in Sec. 3.1, we build our deformation encoder by first extracting a 3D point cloud from the mesh. However, in this

case we are interested in encoding an *animated mesh sequence* $\{\mathcal{M}_t\}_{t=1}^T$. We assume that meshes in the sequence *correspond*, meaning that the i-th vertex in each mesh corresponds to the i-th vertex in every other mesh. We first extract a point cloud $\mathcal{P}_1$ by sampling mesh $\mathcal{M}_1$ as before. We then express each 3D point in terms of $\mathcal{V}_1$ by finding its barycentric coordinates with respect to the face it belongs to, and determine its corresponding location at time t by reconstructing it from vertices $\mathcal{V}_t$. With this, we build a sequence of point clouds $\{\mathcal{P}_t\}_{t=1}^T$ that correspond to each other across time.

These points and corresponding normal vectors are combined and projected to a higher dimensionality:

$$\boldsymbol{h}_t = f_l(\mathrm{PE}(\mathcal{P}_1) \oplus \boldsymbol{n}_1 \oplus \mathrm{PE}(\mathcal{P}_t) \oplus \boldsymbol{n}_t), \tag{3}$$

where PE is positional embedding, $\oplus$ is the channel-wise concatenation operation (which makes sense because points correspond over time), and f_l is a linear layer. By pairing points at time 1 and t, we help the model figure out the motion of the object; by passing the normals, we further help it to determine the local motion. Overall, we obtain a sequence of point features $\{\boldsymbol{h}_t\}_{t=1}^T$, where $\boldsymbol{h}_t \in \mathbb{R}^{M\times c}$ and c is the feature dimension.

Injecting skeleton information. Note that the VAE encoder takes in more information than the decoder outputs. The encoder takes as input the full mesh sequence $\{\mathcal{M}_t\}_{t=1}^T$, while the decoder only outputs the displacement $\mathcal{T}_{1\to t}(\mathcal{V}_1)$ of the vertices. In fact, the encoder takes as input also the $\boldsymbol{n}$, skinning weights $\boldsymbol{w}$ and bones $\boldsymbol{b}$. This can be considered 'privileged' information available at training time to learn a better latent space, but not used at inference time, where the latent code is output by a generator from the monocular video. While points and normals have been used before for a similar purpose (e.g., in VecSet), here we propose to use the skeleton of the object too. This is captured by the *skinning weights* $\boldsymbol{w} \in \mathbb{R}^{M\times B_{\text{max}}}$ that represent the influence of each bone on each point, where B_{max} is the maximum number of bones (we set $B_{\text{max}} = 64$ and unused bones simply get zero weights). To account for the skinning weights, we apply self-attention to the point features $\boldsymbol{h}_t$ with a bias that depends on skinning:

$$\hat{\boldsymbol{h}}_t = \mathrm{softmax}\left(\frac{\boldsymbol{h}_t \boldsymbol{h}_t^\top + M^s}{\sqrt{c}}\right)\boldsymbol{h}_t + \boldsymbol{h}_t$$
$$M^s = \begin{cases} 0 & \text{if } \boldsymbol{w}\boldsymbol{w}^\top > \tau^s \\ -\inf & \text{if } \boldsymbol{w}\boldsymbol{w}^\top \le \tau^s \end{cases}, \tag{4}$$

where c is the channel dimension and $M^s \in \mathbb{R}^{M\times M}$ is the attention mask, which is calculated based on the similarity of the skinning weights and the threshold τ^s.

In addition to biasing the self-attention layer, we incorporate the bone information via cross-attention. Each bone is first represented by the position of its head and tail in each frame, captured by matrices $\boldsymbol{b}_t^h, \boldsymbol{b}_t^t \in \mathbb{R}^{B_{\text{max}}\times 3}$. Again, the unused bones are padded with zeros. The bone parameters are then mapped to the same feature dimension as the point features $\boldsymbol{h}_t$ via a linear layer $\boldsymbol{h}_t^b = f_l^b(\mathrm{PE}(\boldsymbol{b}_1^t) \oplus \mathrm{PE}(\boldsymbol{b}_1^h) \oplus \mathrm{PE}(\boldsymbol{b}_t^t) \oplus \mathrm{PE}(\boldsymbol{b}_t^h))$, $\boldsymbol{h}_t^b \in \mathbb{R}^{B_{\text{max}}\times c}$. Then, the point features attend these bone features via cross-attention:

$$\boldsymbol{h}_t' = \mathrm{softmax}\left(\frac{\hat{\boldsymbol{h}}_t \boldsymbol{h}_t^{b\top} + M^b}{\sqrt{c}}\right)\boldsymbol{h}_t^b + \hat{\boldsymbol{h}}_t$$
$$M^b = \begin{cases} 0 & \text{if } \boldsymbol{w} > \tau^b \\ -\inf & \text{if } \boldsymbol{w} \le \tau^b \end{cases}, \tag{5}$$

where $M^b \in \mathbb{R}^{M\times B_{\text{max}}}$ is the attention mask calculated based on the skinning weights. In Tab. 3, we show that incorporating the skeleton information in the encoder brings significant gain.

Spatio-temporal attention. Rather than modeling the motion of each frame independently [5], we use a transformer [50] with alternating spatial, temporal, and global attention layers [51] to allow the model to correlate the trajectories of different points on the mesh across the frames.

However, the computational cost of applying attention on all points across all frames is prohibitive. Hence, we further downsample the points by applying Farthest Point Sampling (FPS), obtaining $\boldsymbol{h}_t'' \in \mathbb{R}^{N\times c}$ and performing cross-attention with $\boldsymbol{h}_t' \in \mathbb{R}^{M\times c}$ to obtain a sparser set of latent vectors $\{\boldsymbol{h}_t^0\}_{t=1}^T$, $\boldsymbol{h}_t^0 \in \mathbb{R}^{N\times c}$,

$$\boldsymbol{h}_t^0 = \mathrm{softmax}\left(\frac{\boldsymbol{h}_t'' \boldsymbol{h}_t'^\top}{\sqrt{c}}\right)\boldsymbol{h}_t', \tag{6}$$

where $N \ll M$ and c is the channel dimension. Then, as shown in Fig. 2 (right), we apply spatial self-attention to the tokens $\boldsymbol{h}_t^l$ within each frame separately, temporal self-attention to the tokens across all frames but separate points, and global attention to all tokens $\cup_{t=1}^T\{\boldsymbol{h}_t^l\}$ across all frames jointly. These attention layers are interleaved with MLP layers to further enhance the representation.

After $L = 8$ layers of alternating attention, an additional linear projection layer is applied to obtain the mean $\mathrm{E}(\boldsymbol{h}_t^L) \in \mathbb{R}^{N\times c_o}$ and variance $\mathrm{Var}(\boldsymbol{h}_t^L) \in \mathbb{R}^{N\times c_o}$ of the deformation latent distribution for each frame t, where $\boldsymbol{h}_t^L$ is the output of the last attention layer L for frame t, and $c_o \ll c$. Finally, the latent $\boldsymbol{z}^d$ output by the encoder is sampled from $\mathcal{N}(\mathrm{E}(\boldsymbol{h}^L), \mathrm{Var}(\boldsymbol{h}^L))$.

Deformation field decoder. The decoder $\mathcal{T}_{1\to t}(\mathcal{V}_1) = \mathcal{D}_t^d(\mathcal{V}_1; \boldsymbol{z}^d)$ recovers the deformation field in correspondence of the mesh vertices from the latent code $\boldsymbol{z}^d$ output by the encoder. First, we use a linear layer to project $\boldsymbol{z}^d$ from dimension c_o to a higher dimension c. Then, we apply 16 layers of spatio-temporal attention blocks as described

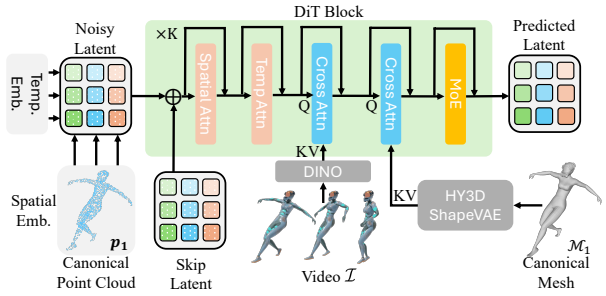

Figure 3. **Overall deformation diffusion model pipeline.** We build it based on HY3D 2.1 [46] shape diffusion model with additional spatial and temporal embedding as well as cross attention layer to condition the deformation field generation on the canonical mesh and input video.

in Fig. 2 to further enhance the feature representation. Finally, the vertices $\mathcal{V}_1$ are used as queries to recover the deformation field $\mathcal{T}_{1\to t}(\mathcal{V}_1)$ via cross attention.

Training objective. The deformation VAE is trained by minimizing the following loss:

$$\mathcal{L}_{\text{VAE}} = \sum_{t=1}^{T} \left\| (\mathcal{V}_t - \mathcal{V}_1) - \mathcal{D}_t^d(\mathcal{V}_1; \boldsymbol{z}^d) \right\|_2^2 + \lambda L_{\text{KL}},$$
$$\boldsymbol{z}^d \sim \mathcal{E}^d(\{\mathcal{M}_t\}_{t=1}^T; \boldsymbol{n}, \boldsymbol{w}, \boldsymbol{b}). \quad (7)$$

L_{KL} is a KL divergence loss that regularizes the latent space. In practice, we evaluate this loss only for a random subset of vertices for efficiency.

3.3. Deformation diffusion model

As shown in Fig. 3, given the reconstructed canonical mesh $\mathcal{M}_1$ and the input video $\mathcal{I}$, we use a diffusion model to generate the deformation latent $\boldsymbol{z}^d \sim p(\boldsymbol{z}^d|\mathcal{M}_1, \mathcal{I})$ conditioned on both signals. To build the deformation diffusion model, we extend the shape diffusion model in Sec. 3.1 so that it learns a velocity field $\boldsymbol{v}_{\boldsymbol{\theta}}^d$ with additional temporal embeddings and spatial embeddings $\boldsymbol{p}_1 \in \mathbb{R}^{N\times 3}$ sampled from $\mathcal{M}_1$, as well as extra attention layers in each DiT block to fuse temporal video cues and canonical shape information. Video features are extracted from DINO-Giant [36] frame by frame and cross-attended by the corresponding latent tokens. Similar to GVFD [60], the spatial embedding $\boldsymbol{p}_1$ endows the initial noise with spatial awareness, which improves spatial consistency. During training, we use the positions of the FPS-sampled sparse features $\boldsymbol{h}''$ as spatial embeddings to align each latent target with its embedding. During inference, we perform FPS sampling on the reconstructed canonical mesh to obtain the required spatial embedding. Different from GVFD [60], we also incorporate temporal embeddings for better temporal consistency and leverage the extracted high-dimensional shape feature $\boldsymbol{z}^s$ from the canonical mesh for detailed conditioning. The training objective matches Eq. (2).

Method	Reconstruction			Tracking
	IoU ↑	P2S ↓	Chamfer ↓	ℓ_2-Corr ↓
HY3D 2.1 [46]	0.3071	0.0376	0.0370	N/A
L4GM [41]	N/A	0.0459	0.0505	N/A
GVFD [73]	N/A	0.0345	0.0378	0.0514
Ours	0.3731	0.0287	0.0273	0.0384
Ours (Aligned)	**0.3949**	**0.0261**	**0.0243**	**0.0338**

Table 1. **Quantitative evaluation** for geometry and tracking on our proposed benchmark, a subset of Objaverse. All instantiations of our model outperform prior state-of-the-art methods. 3D-GS based methods do not explicitly define inner or outer surfaces, so volumetric IoU is not applicable. HY3D [46] and L4GM [41] predict independent meshes or points per frame, so they do not directly support tracking evaluation.

4. Experiments

4.1. Experimental settings

Dataset. We start from the curated version of Objaverse-1.0 [9] released by Diffusion4D [28], where objects animated with limited motion or large distortions are filtered out. We extract the skeleton, skinning weights, and the sequence of meshes with corresponding vertices. We then filter out objects with an excessively large number of vertices or bones, retaining approximately 9k instances. Each instance is rendered as a frontal video with up to 100 frames. For testing, we select a disjoint set of 50 sequences with significant object motion and high-quality textures. We render four fixed-view videos for each sequence at azimuth angles $0°, 90°, 180°, 270°$. One view is used as the input and the other three for novel view synthesis (NVS) evaluation.

Baselines. We compare our method with three state-of-the-art latent 3D reconstruction models: Hunyuan3D 2.1 (HY3D) [46], L4GM [41], and GVFD [73]. Since HY3D is an image-to-3D reconstruction method, we run it frame by frame with shared noise to improve temporal consistency. For L4GM and GVFD, we take the centers of Gaussian primitives with opacity greater than 0.01 as a point cloud for geometry evaluation. To align the reconstructed mesh with the ground-truth mesh, we perform Coherent Point Drift (CPD) [34] to optimize the scale and rigid transformation of the first-frame mesh. As shown in Tabs. 1 and 2, entries with the *Aligned* annotation indicate that the aligned canonical shape is used as input to our deformation diffusion model, whereas entries without this annotation indicate that alignment is performed immediately before evaluation.

Metrics. For frame-wise *geometry* evaluation we report volumetric IoU, point-to-surface distance (P2S), and Chamfer distance. For *tracking*, we measure the Euclidean distance between corresponding points (nearest neighbors on the first frame) on the predicted and ground-truth meshes (ℓ_2-Corr). For *novel view synthesis (NVS)* [7, 76], we report

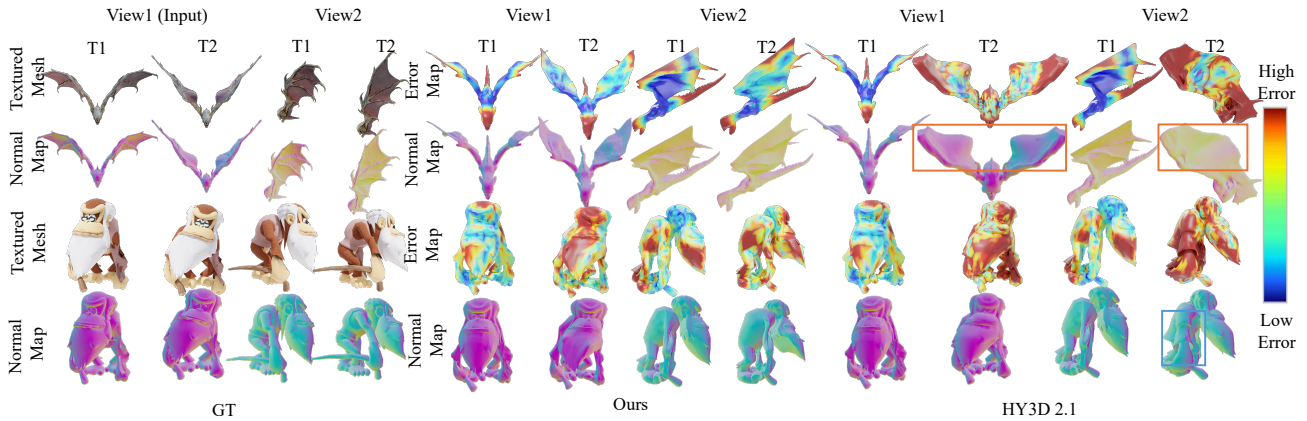

Figure 4. **Qualitative results on geometry reconstruction.** We show both the normal map and the error map, with bluer tones indicating lower error. HY3D 2.1 [46] suffers from inaccurate **pose** and **shape** estimation because it lacks temporal information. Thanks to spatio-temporal attention, our method reconstructs meshes that follow the input frames with accurate pose and similar shape.

Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), Learned Perceptual Image Patch Similarity (LPIPS), and CLIP similarity. We also report FVD [49] to assess the temporal consistency of generated novel-view videos.

4.2. Geometry and tracking evaluation

In Fig. 4 we qualitatively compare our method with frame-wise HY3D inference [46]. Due to the lack of temporal attention across frames, HY3D suffers from inaccurate pose (incorrect pose of the chimpanzee's arm in the **blue** box) and shape (incorrect bat wing shape in the **orange** box). This is confirmed by the quantitative results in Tab. 1, where Mesh4D achieves state-of-the-art reconstruction and tracking performance by leveraging information across the whole sequence. 3D-GS-based methods do not inherently focus on geometry or dense tracking, which is also reflected in the ghost artifacts (see Fig. 5 discussed next).

4.3. Novel view synthesis evaluation

In Fig. 5 we qualitatively assess NVS results on both the input view and a novel view at different time steps. Typical errors fall into four categories, described below.

Inaccurate pose estimation. As shown in the **blue** boxes of Fig. 5, HY3D predicts inaccurate poses (e.g., human legs, mantis limb). In contrast, our method predicts pixel-aligned pose estimates via temporal and global attention, yielding better PSNR, SSIM, and LPIPS (see Tab. 2).

Inconsistent texture. Although using shared noise improves HY3D frame-to-frame consistency, it still exhibits jittery geometry and texture flicker (see the **orange** boxes). Our model produces consistent texture and geometry thanks to dense correspondences modeled by the spatio-temporal attention, leading to a lower FVD (Tab. 2).

Method	Per-frame Quality				Consistency
	PSNR ↑	SSIM ↑	LPIPS ↓	CLIP ↑	FVD ↓
HY3D 2.1 [46]	19.14	0.8976	0.1195	**0.9174**	692.2
L4GM [41]	18.07	0.8939	0.1453	0.8954	747.3
GVFD [73]	17.31	0.8912	0.1459	0.8802	905.0
Ours	19.67	0.9018	0.1087	0.9141	601.9
Ours (Aligned)	**19.88**	**0.9030**	**0.1052**	0.9141	**572.7**

Table 2. **Quantitative evaluation** for novel view synthesis on our proposed benchmark, a subset of Objaverse. We achieve the best performance on both frame-wise quality and video consistency.

Method	Reconstruction			Tracking
	IoU ↑	P2S ↓	Chamfer ↓	ℓ_2-Corr ↓
w/o temp&global attention	0.6328	0.0153	0.0113	0.0160
w/o skeleton information	0.6704	0.0148	0.0107	0.0138
Ours	**0.7039**	**0.0144**	**0.0099**	**0.0117**

Table 3. **Quantitative ablation study** for our deformation VAE on our proposed benchmark, a subset of Objaverse. We demonstrate the effectiveness of our key designs.

Incorrect canonical mesh. Our method builds upon a large-scale 3D reconstruction model that produces reasonably accurate canonical shapes for diverse objects. On the contrary, GVFD and L4GM often fail to recover an accurate canonical shape, a prerequisite for high-quality 4D reconstruction (**black** boxes in Fig. 5).

Artifacts of deformed 3D-GS. Prior methods that generate 3D-GS representations rely on photometric losses without shape and motion priors. They suffer from ghost artifacts under large motion (**red** box in Fig. 5).

Our texture map is generated from only the first input frame, so our method yields a lower CLIP score compared to HY3D, which generates a new texture for each frame. However, thanks to the reconstructed deformation field, we estimate more accurate motion and alignment, achieving state-of-the-art results on all other metrics (Tab. 2).

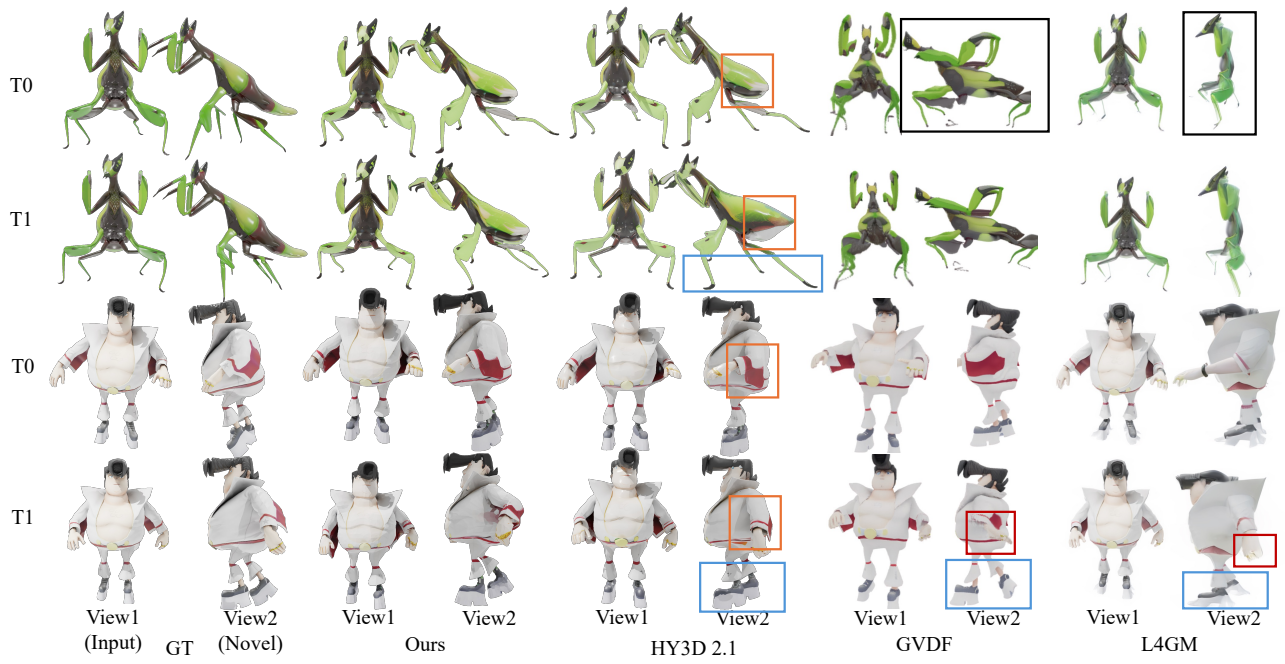

Figure 5. **Qualitative results on novel view synthesis.** All state-of-the-art methods suffer from **inaccurate pose estimation**, either because they lack temporal attention (HY3D [46]) or because they neglect geometric supervision (GVDF [73], L4GM [41]). 3D-GS based methods occasionally exhibit **ghost artifacts** because they lack topology constraints during deformation, while frame-wise reconstruction methods produce **inconsistent shape and texture**. Leveraging a large reconstruction model prevents extremely **incorrect canonical meshes**. Thanks to the skeleton information and spatio-temporal attention, Mesh4D reconstructs accurate pose and geometry and produces temporally consistent novel-view videos.

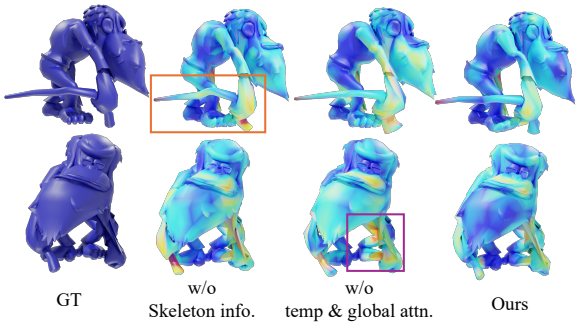

Figure 6. **Ablating key components of the deformation VAE.** We visualize the error map of the chamfer distance. Blue indicates lower error, corresponding to better reconstruction quality. Injecting skeleton information helps the model better capture rigid deformation. Spatio-temporal fusion reduces jittering effects.

4.4. Ablation study

We ablate key design choices of the deformation VAE. We train two variants: one without skeleton information and one without temporal and global attention. At test time we use the ground-truth first-frame mesh as the canonical mesh and evaluate the reconstructed deformation field. As shown in the **orange** box of Fig. 6, removing skeleton information impairs rigid transformations, resulting in a twisted stick. As shown in the **purple** box, removing temporal and global attention causes jittery motion and larger errors near the feet. These observations align with the quantitative results in Tab. 3, confirming the effectiveness of our VAE design.

5. Conclusion

We introduced Mesh4D, a feed-forward model for 4D mesh reconstruction from monocular videos. Starting from a latent 3D reconstruction model pre-trained on a large collection of static objects, we add a new VAE that encodes the object deformation in a compact latent space. We further introduce a method that uses skeleton information to supervise this VAE and a diffusion model built on top of these components. Thanks to this design, Mesh4D predicts the full 3D shape of the object as well as its deformation, tracking vertices throughout the entire video sequence.

On the Objaverse benchmark, Mesh4D achieves state-of-the-art reconstruction quality for geometry, motion, and appearance, while reducing temporal artifacts relative to competing approaches. Our ablations highlight the value of our contributions, including using skeletal cues and spatio-temporal attention in the VAE architecture. Limitations include reliance on a high-quality canonical mesh and skeletons for training, the inability to represent topological changes in the mesh, and difficulty in reconstructing very non-rigid objects. Code, models, benchmark splits, and the full evaluation framework are available.

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